Application Commands — Cheatsheet

Every command to run the app, by what you want to do.

Run the whole thing

```bash
python demo.py
```

The menu-driven presentation CLI. Pick pipeline / semantic search / RAG / ask-your-own from the menu. **This is the one for Thursday.**

Jump straight to a section:

```bash
python demo.py --pipeline      # ETL pipeline only
python demo.py --search        # semantic search only
python demo.py --rag           # RAG only
python demo.py --ask           # live "ask your own question" mode
```

The ETL pipeline

```bash
python main.py                 # extract → load → export → upload
python main.py --demo          # the above, then a CRUD walkthrough
```

Semantic search

```bash
python embed_and_search.py --embed        # generate embeddings — RUN ONCE
python embed_and_search.py --demo         # the 4-query demo
python embed_and_search.py "electric cars"    # your own query
```

Examples that land well:

```bash
```

```
python embed_and_search.py "companies that make computer chips"
python embed_and_search.py "streaming services"
python embed_and_search.py "cloud software"
```

RAG (ask in plain English)

```bash
python ask.py --demo                                        # the demo questions
python ask.py "which companies make chips and how are they doing?"
python ask.py "is any streaming company down today?"
python ask.py "tell me about the EV company"
```

Tests

```bash
pytest -v                           # all 24
pytest tests/test_search.py -v        # just semantic search
pytest tests/test_rag.py -v           # just RAG
```

Inspect the data (mongosh)

```bash
mongosh "mongodb://etladmin:<pw>@<EC2_IP>:27017/?authSource=admin"
```

```javascript
use stockdb
db.companies.countDocuments({})                             // total companies
db.companies.countDocuments({ embedding: {$exists:1} })     // how many embedded (
db.companies.findOne()                                      // one full document
db.companies.find({ industry: "Semiconductors" })           // filter by industry
```

The typical order, start to finish

```bash
```

```
python main.py                            # 1. pull data → Mongo → S3
python embed_and_search.py --embed        # 2. add embeddings (once)
python ask.py --demo                      # 3. warm the RAG cache
python demo.py                            # 4. present
```

Two things that must be done ONCE, ahead of time

```bash
python embed_and_search.py --embed        # downloads the model (~90MB), then off
python ask.py --demo                      # populates rag_cache.json (offline fal
```

Do both on a network you trust — **not** five minutes before presenting.

Pipeline: `Finnhub → Python → MongoDB (EC2) → JSON → S3 → Semantic Search → RAG`